An Efficient Context Management System for On-Device LLMaaS

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On-Device Large Language Models as a Service (LLMaaS)

Apple Intelligence

Single Instance

Many Apps

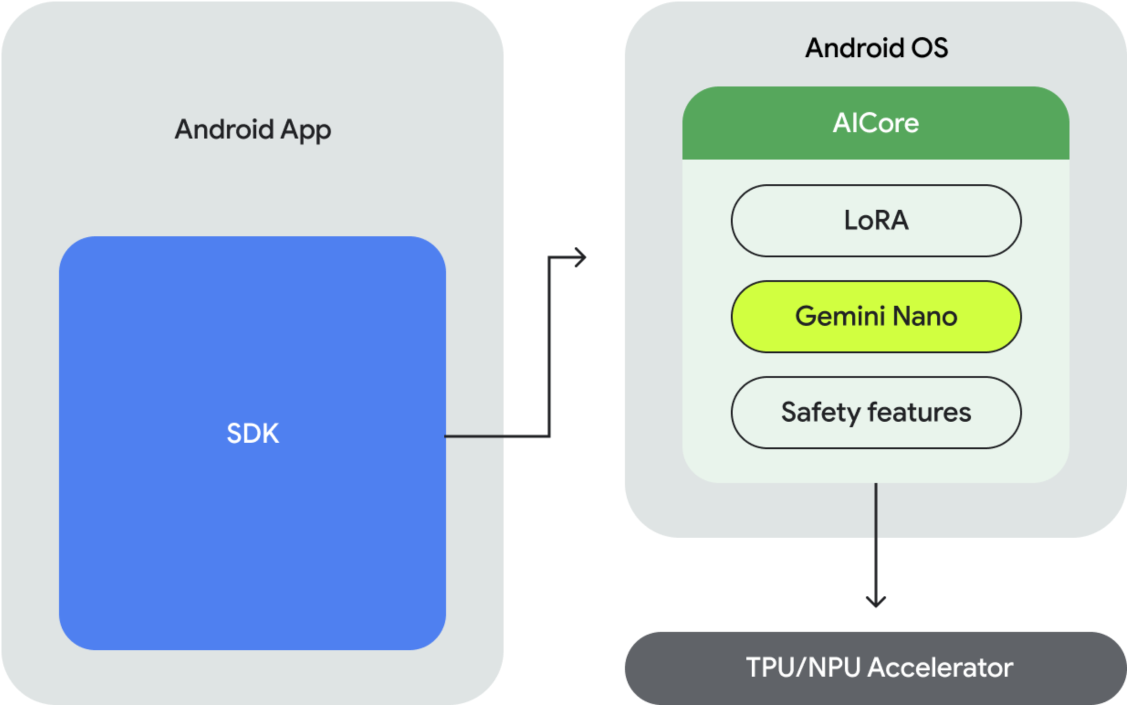
Text Gen

Emoji Gen

Hey Siri

Private, and efficient

On-Device Large Language Models as a Service (LLMaaS)



Google Android AICore

Has been integrated into Android as *com.google.ai.edge.aicore*

https://developer.android.com/ai/reference/com/google/ai/edge/aicore/package-summary

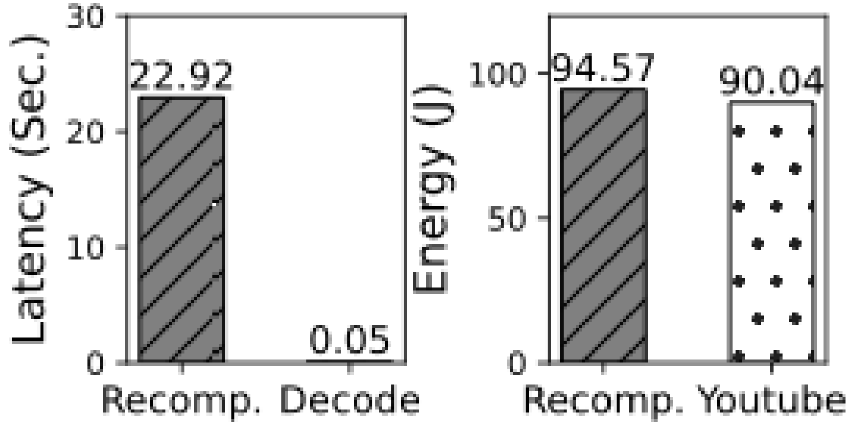
The Key Problem: Stateful On-Device LLMaaS

Asking Siri: What is the restaurant that you recommended yesterday?

Google keyboard: complete current text by historical information.

The Key Problem: Stateful On-Device LLMaaS

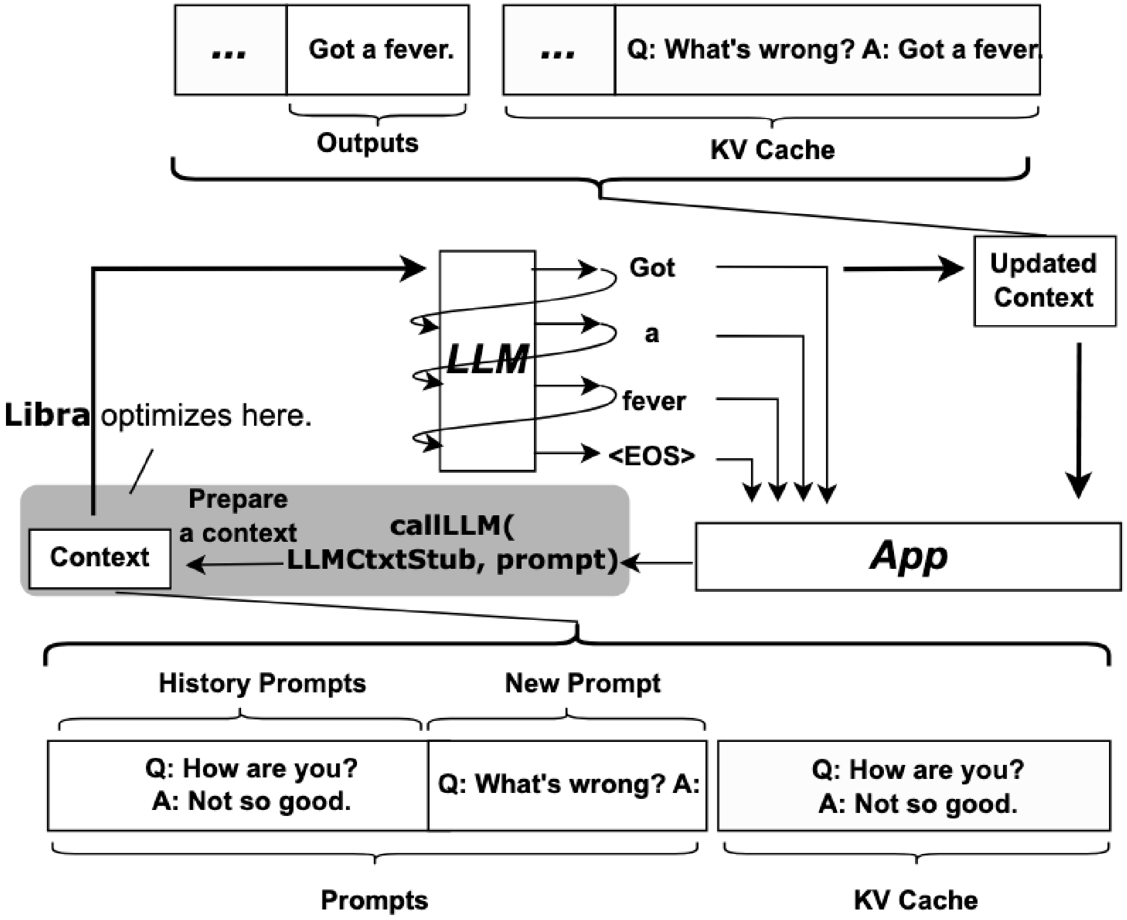
Keeping the state of LLM service is with heavy overhead…



KV cache memory dominates as the contexts (apps/sessions) grow.

Delegate the context to OS (e.g. android low memory killer) and recompute the KV cache? That's costly!

Libra: Context Management System for On-Device LLMaaS

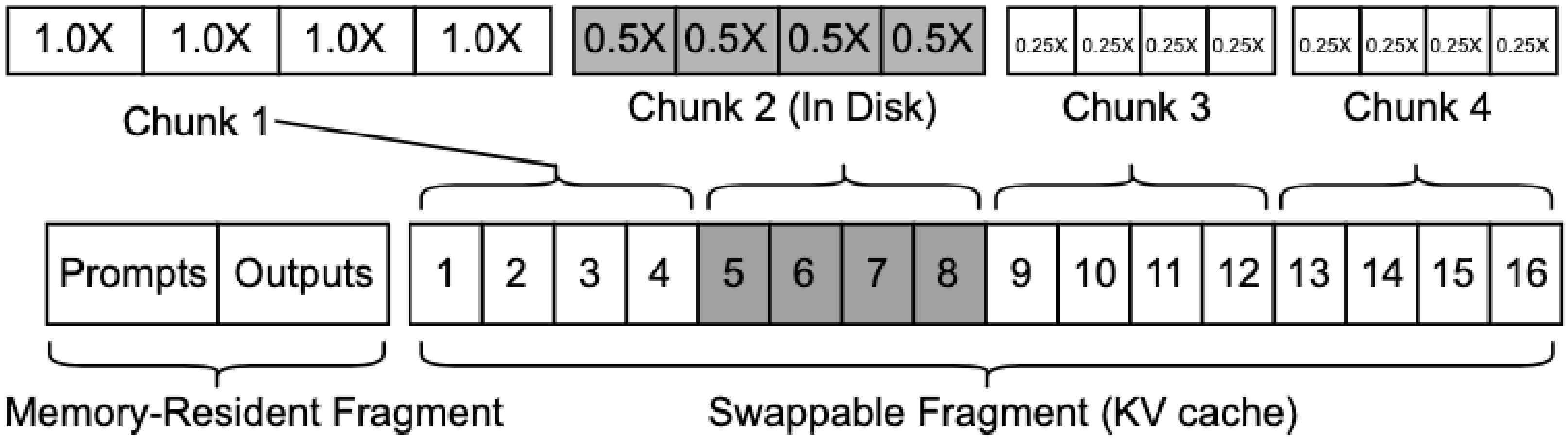



```java
public class ChatBot extends AppCompatActivity
    {
  /* App codes */
  ...
  private void newSession(){
    // bind LLM service to current app
    LLMService llmService =
      bindLLMService(curApp);
    // new LLM context
    LLMCtxStub llmCtx = newLLMCtx();
    // chatbot logic
    while(prompt = getText()){
      String result =
        llmService.callLLM(llmCtx, prompt);
      Log.d(result);
    }
    // delete LLM context when session ends
    delLLMCtx(llmCtx);
  }}
```

The Goal: Fast Context Preparation

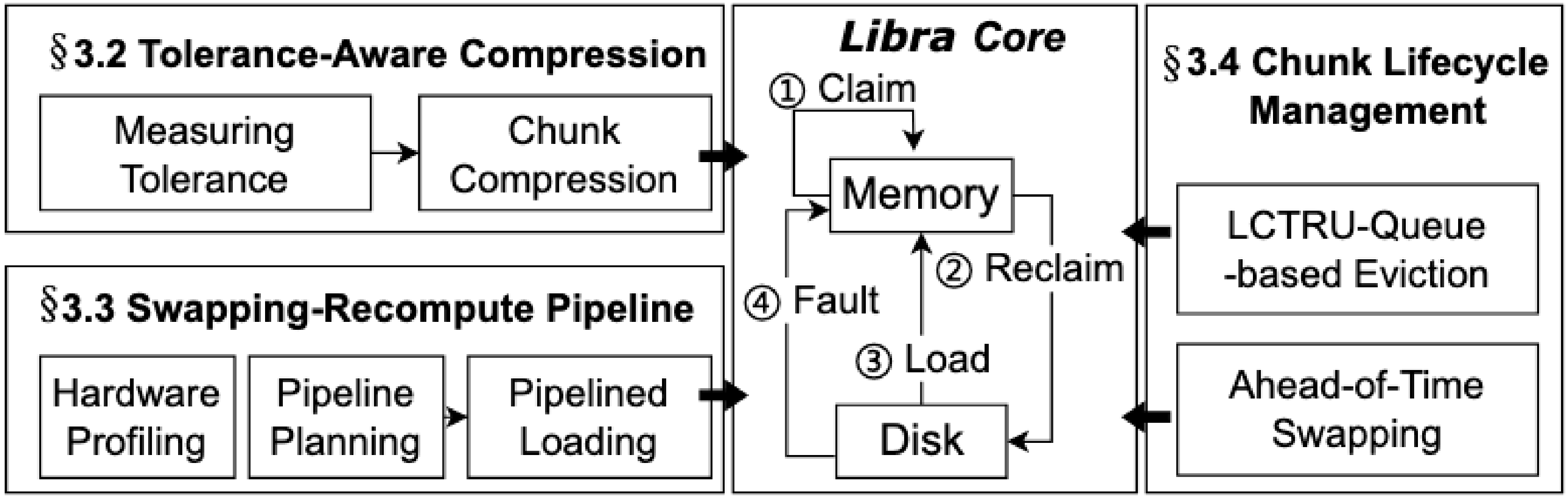
The Interface: OS-Aligned APIs

The Idea: Flexible Chunks



- **Chunk in groups tokens**
- **Each chunk has its own compression ratio**
- **Each chunk can be swapped in/out arbitrarily**

Main Techniques at a Glance



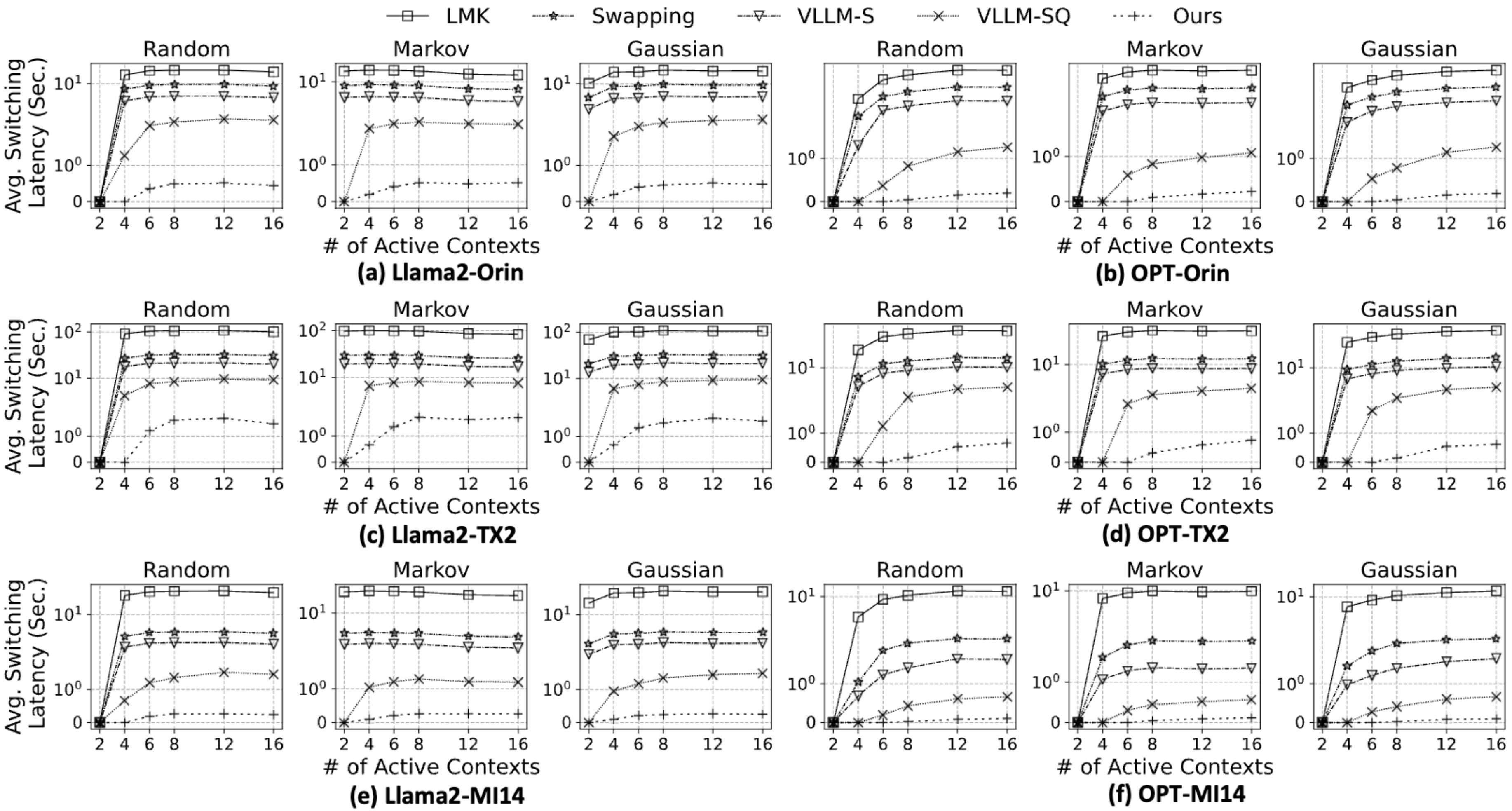
- **Technique#1 Tolerance-Aware Chunk Compression**
  Determine the compression ratio by chunk semantics

- **Technique#2 Swapping-Recompute Pipeline**
  Swapping and recompute in parallel for disk input acceleration

- **Technique#3 Chunk Lifecycle Management**
  Select proper swapping timing and priority for each chunk

See our detailed insights in the paper!

The Evaluation



Up to 20x, on average 9.7x context switching latency reduction on a 72-hours-long synthetic trace.

The Evaluation

Dataset (Metric)	AGNews (Acc.)		Xsum (RG-L)		Samsum (RG-L)		CNNDM (RG-L)		WMT17 (RG-L)		SST2 (Acc.)		MMLU (Acc.)		DroidTask (Step Acc.)		Avg.	
Models/Methods	**Llama2**	**OPT**	**Llama2**	**OPT**	**Llama2**	**OPT**	**Llama2**	**OPT**	**Llama2**	**OPT**	**Llama2**	**OPT**	**Llama2**	**OPT**	**Llama2**	**OPT**	**Llama2**	**OPT**
16bits ※	**59.0**	**24.5**	**13.1**	**12.3**	**18.9**	**13.7**	**22.9**	**24.3**	**34.4**	**31.2**	**88.7**	**63.8**	**37.2**	**28.8**	**14.4**	**17.4**	***36.1***	***27.0***
Static-8 ※	**58.7**	**24.7**	**13.1**	**12.2**	**18.8**	**12.2**	**22.8**	**24.3**	**34.2**	**31.2**	**88.8**	**63.8**	**37.1**	**28.7**	**13.9**	**16.8**	***36.0***	***26.7***
Static-4	37.6	23.7	13.3	11.9	16.2	10.7	19.7	20.8	28.7	27.3	63.2	60.2	32.4	23.7	11.4	12.6	27.8	23.9
Static-2	25.3	23.2	1.1	2.7	2.2	1.7	0.9	3.7	5.9	8.7	50.0	51.0	25.1	24.2	6.7	9.8	14.7	15.6
Libra(50%) ※	**58.7**	**25.2**	**12.7**	**12.1**	**18.2**	**12.2**	**22.0**	**22.8**	**33.3**	**30.8**	**88.7**	**63.9**	**36.7**	**28.0**	**14.8**	**18.0**	***35.6***	***26.6***
Libra(35%)	54.2	23.6	10.3	11.5	16.9	11.7	19.2	19.1	28.1	29.9	83.2	58.7	31.1	24.7	8.7	14.7	31.5	24.2

Visualized average compression ratio ※Methods with minimal quality loss (<1% on average) compared to FP16

Almost lossless in accuracy on various datasets against strong compression baselines.

Take Aways

- **We present Libra, the first-of-its-kind context managing system for on-device LLMaaS.**

- **Exploiting the idea of chunking, it incorporates several fancy and penetrating techniques.**

- **Impressive reduction on context switching latency, with lossless accuracy drop.**